राष्ट्रीय भू-सूचना विज्ञान एवं प्रौद्योगिकी संस्थान
भारतीय सर्वेक्षण विभाग
विज्ञान और प्रौद्योगिकी विभाग

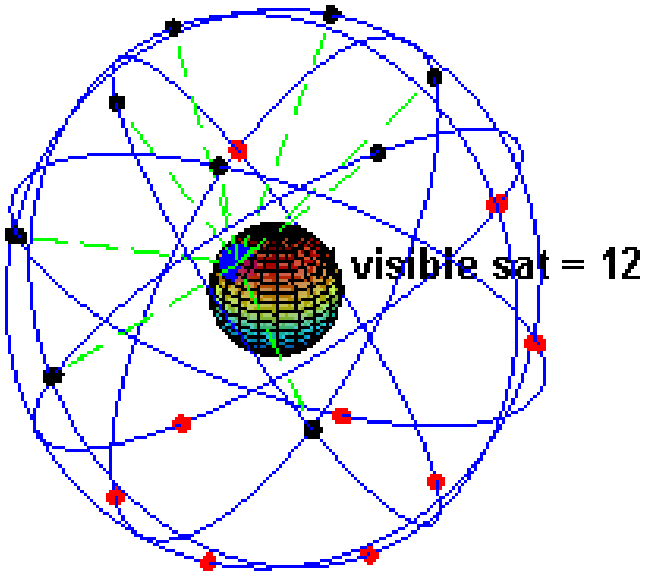
National Institute for Geo-Informatics Science & Technology
Survey of India
Department of Science & Technology

Route Planning & Transmission Line Survey Using Modern Technologies

Day 11: 20 Jan 2026

Fundamentals of GNSS & Signals

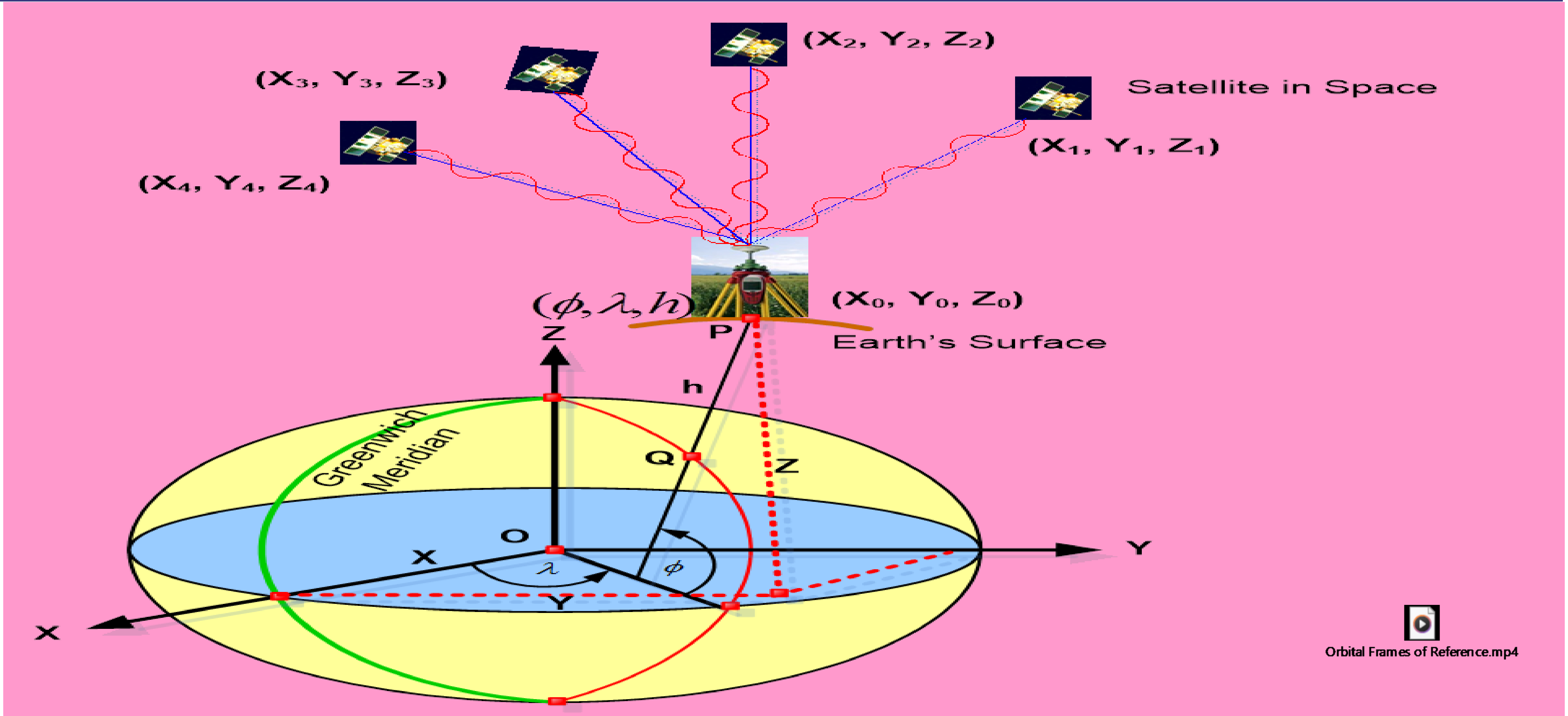
09:15-11:00



Faculty of Geodesy, NIGST, Survey of India

geodesy.nigst@surveyofindia.gov.in

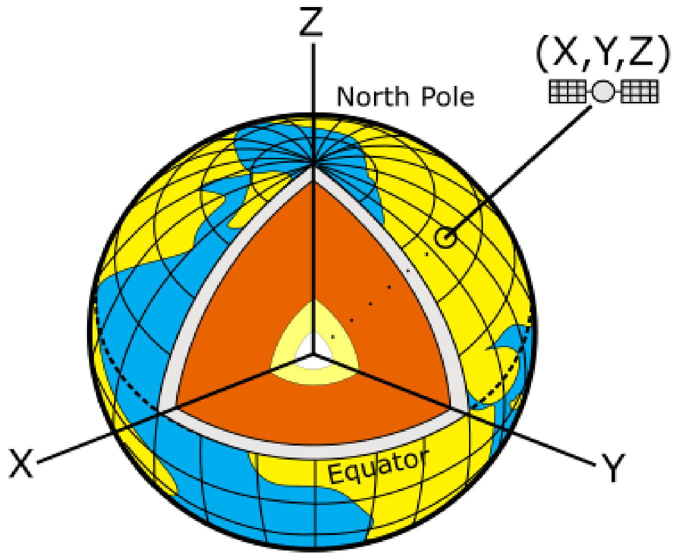
Cartesian coordinates (x, y, z) and Geodetic Coordinates(ϕ,λ,h)

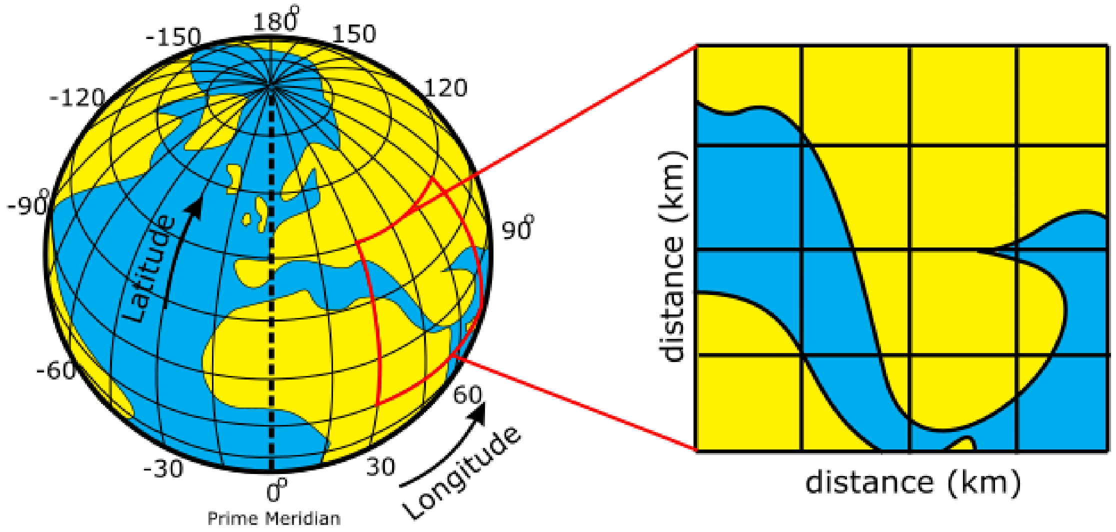


Orbital Frames of Reference.mp4

GNSS provides 3D positioning

- Positions on the earth can be expressed using:
  - Cartesian coordinates (x,y,z w.r.t the earth's center)
  - Geographic coordinates (lat., long., elev.)
  - Projected coordinates (UTM, state plane, in m or ft)





OBSERVATION PRINCIPLE OF GNSS

Satellite position (X_i, Y_i, Z_i) - Known

Receiver Position (X_o, Y_o, Z_o) – Unknown

Then Geometric range(ρ)= $\sqrt{(Xi - Xo)^2 + (Yi - Yo)^2 + (Zi - Zo)^2}$

Signal Transmitted time from satellite – T_t

Signal receiving time by receiver - T_r

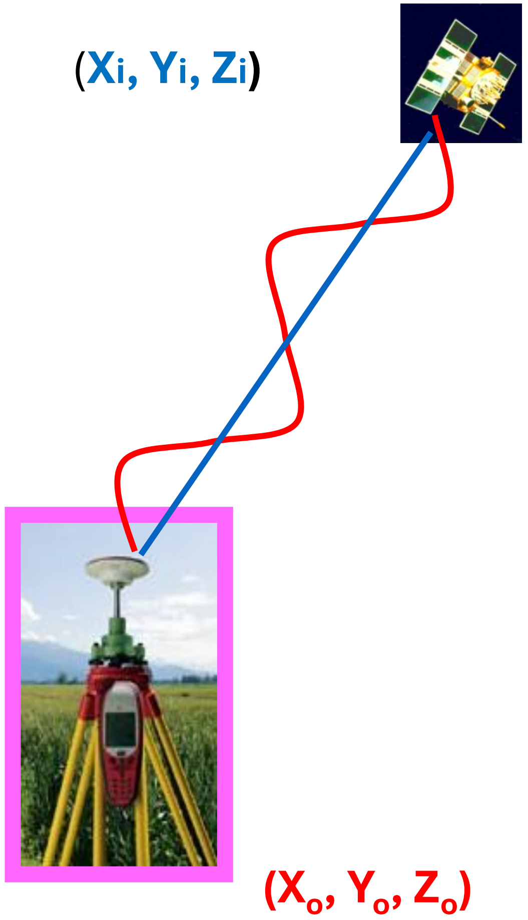
Synchronization error in satellite and receiver clock - T_0

$$\text{Range } R = c\ \{(T_r - T_t) - T_0\}$$

So, $$c\,(T_r - T_t) = \sqrt{(Xi - Xo)^2 + (Yi - Yo)^2 + (Zi - Zo)^2} + c\,T_0$$

Where (X_o, Y_o, Z_o) = **Coordinates of Antenna in ECEF coordinate system**

(X_i, Y_i, Z_i) = **Coordinates of satellites in ECEF coordinate system**



GNSS CONFIGURATION

GPS
- Since 1980s
- 31 Sats
- GPS III Launch Dec 2018

GLONASS
- Refurb 2012
- Operating well with 24 sats

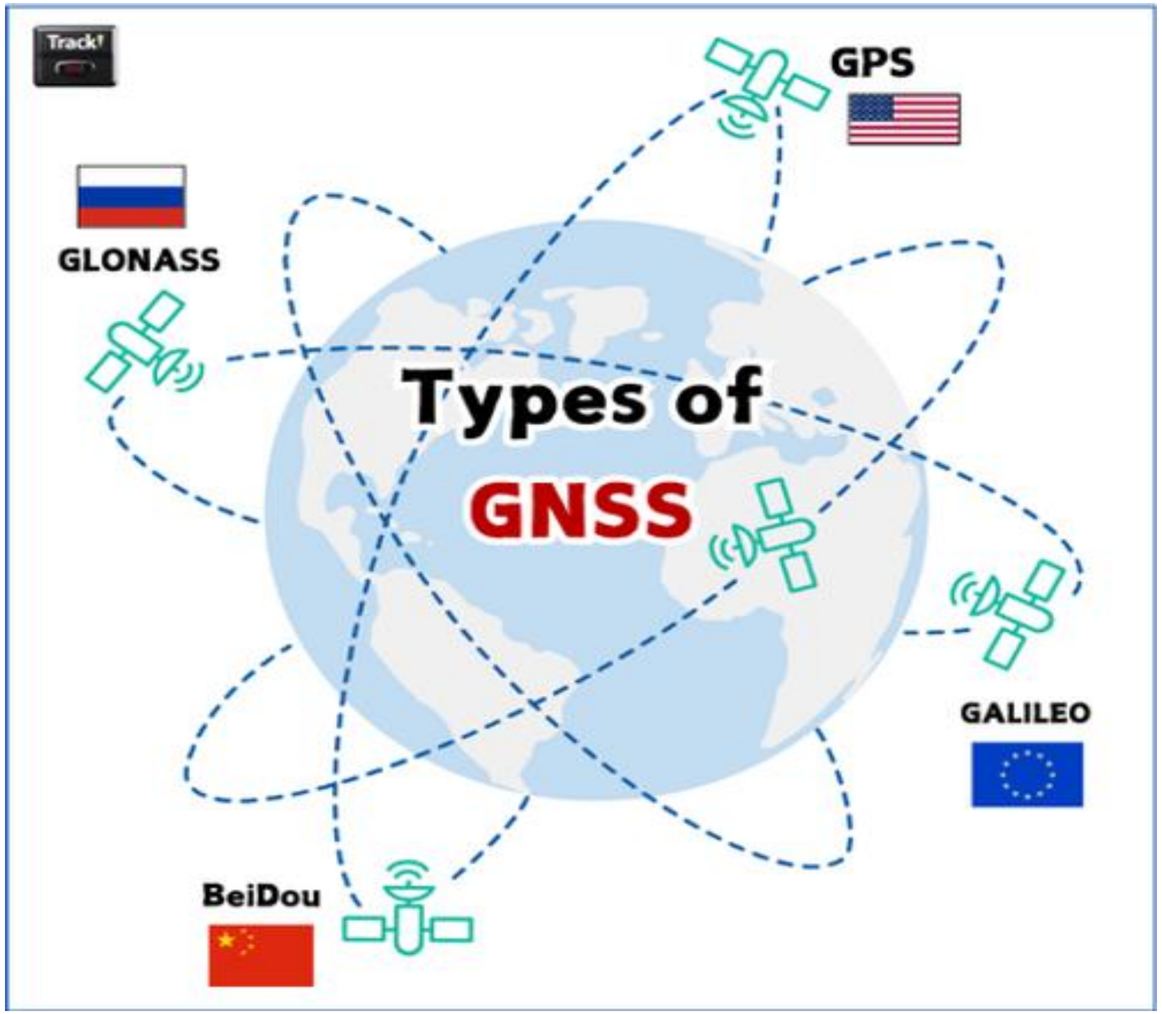
Galileo
- Full capability in 2022
- 30 sats

Beidou
- Full capability in 2020
- 33 sats for BD-3

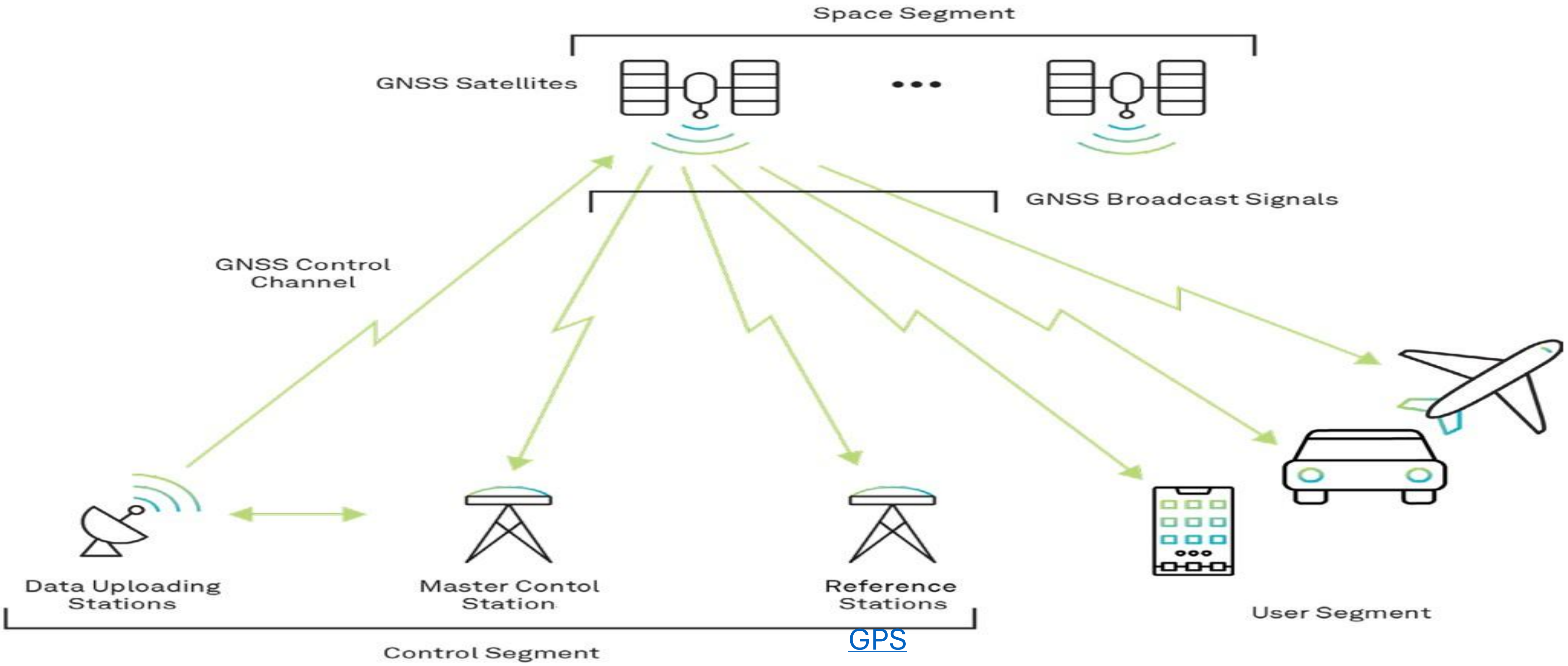
Regional Systems

Japan - QZSS

India - NAVIC



GNSS OVERVIEW



GPS

GNSS Frequencies and Signals (Electromagnetic spectrum)

System	Signal	Frequency (MHz)
GPS	**L1 C/A(154)**	**1575.42**
	L1C	**1575.42**
	L2 C	**1227.6**
	L2 P (120)	**1227.6**
	L5 (115)	**1176.45**
GLONASS	L1 C/A	1598.0625-1609.3125
	L2 C	1242.9375-1251.6875
	L2 P	1242.9375-1251.6875
	L3 OC	1202.025
Galileo	**E1**	**1575.42**
	E5a	**1176.45**
	E5b	**1207.14**
	E5AltBOC	**1191.795**
	E6	**1278.75**

System	Signal	Frequency (MHz)
BeiDou	**B1I**	**1561.098**
	B2I	**1207.14**
	B3I	**1268.52**
	B1C	**1575.42**
	B2a	**1176.45**
	B2b	**1207.14**
NAVIC	L5	1176.45
SBAS	**L1**	**1575.42**
	L5	**1176.45**
QZSS	L1 C/A	1575.42
	L1 C	1575.42
	L1S	1575.42
	L2C	1227.6
	L5	1176.45
	L6	1278.75

GPS Signal Components

Basically, consists of 3 components

Ranging Codes modulated on the Carrier : A unique binary code (C/A or P-code) that helps identify the satellite and measure distance (pseudorange)

Carrier Wave : High frequency radio wave carries the signal, enables very precise measurement via carrier phase

Navigation Message : contains ephemeris data to calculate the position of the satellite in orbit, and information about the time and status of the satellite constellation.

Basic Principle of GNSS

- Trilateration: Knowing the position of GNSS receiver with the help of known positions(distances) of three control points.
- In case of GNSS, the satellites act as control points. They send their location information through the navigation message (in the form of binary codes i.e., 0 and 1) superimposed on the carrier waves.
- In order to find the location correctly, we need to find the distance between the satellite and the receiver (Range) very accurately from at least 4 satellites.
- Distance(Range) is computed by multiplying Velocity with Time of travel.
- The accuracy of the GNSS depends on how accurately the Range is measured.
- The Range can be measured in two ways:

    1. by using the Codes (Code based measurement)
    2. by using the Carrier wave (Phase based measurement)

GPS Signal Structure

- Fundamental Frequency **10.23MHz** (f_0)

3 Carrier Frequencies

- L1 (1575.42 MHz) (154 f_0)
- L2 (1227.60 MHz) (120 f_0)
- L5 (1176.45MHz) (115 f_0)

superimposed with several types of Codes

- Coarse Acquisition (C/A) 1.023 MHz
- Precise (P) 10.23 MHz
- Encrypted (Y)
- Military (M)
- L1C, L2C etc.

on Spread Spectrum.

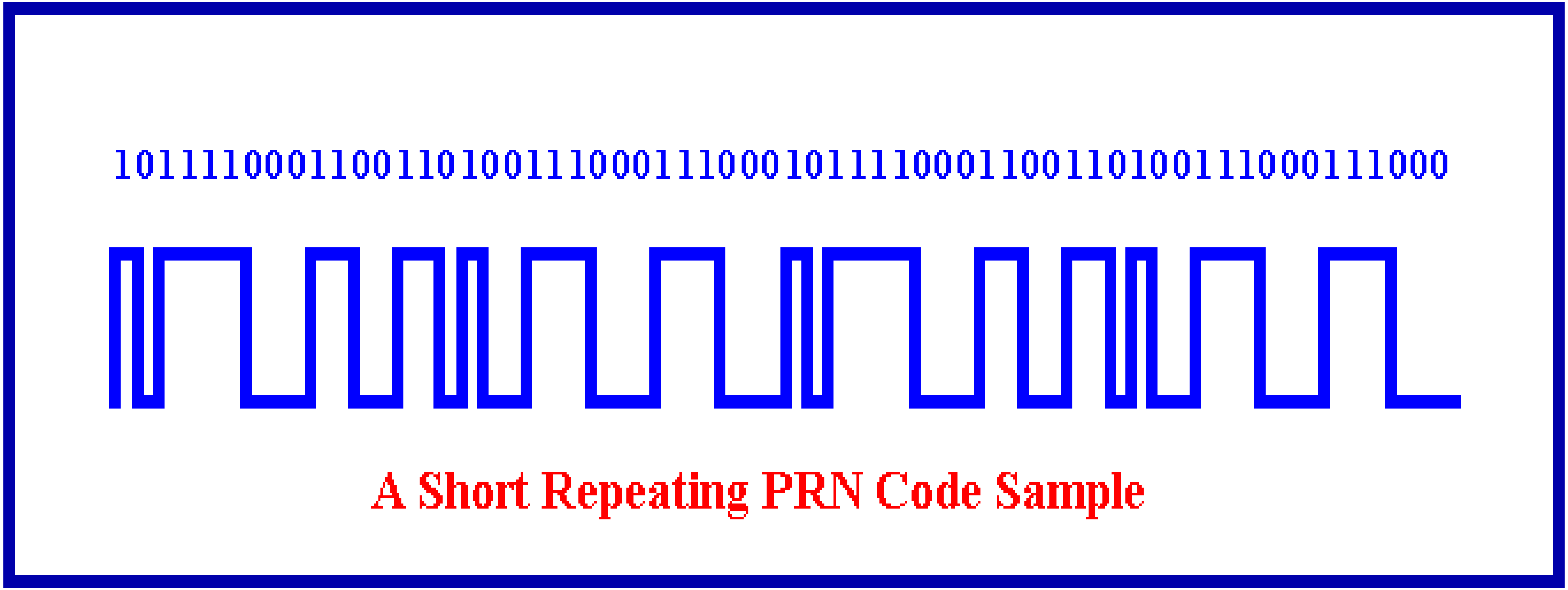
Pseudo-Random Codes

Coarse Acquisition (C/A) Code: A Sequence of 1023 Bi-Phase Modulations of The Carrier, Repeated 1000 x sec. With A "Chip Rate" of 1.023 MHz *(SPS)*

Precise (or Protected) (P) Code: A Very Long Sequence of Bi-Phase Modulations of The Carrier, Repeated every 267 Days With A "Chip Rate" Of 10.23 MHz *(PPS)*

Ranges From These Codes Are Called "Pseudo-Random Ranges"

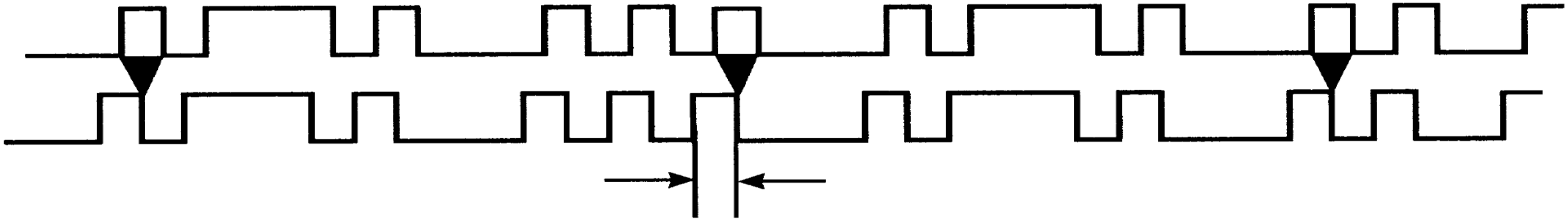
Sample sequence of code



A Short Repeating PRN Code Sample

GPS Resolution - Code Phase

C/A Code 1.023MHz

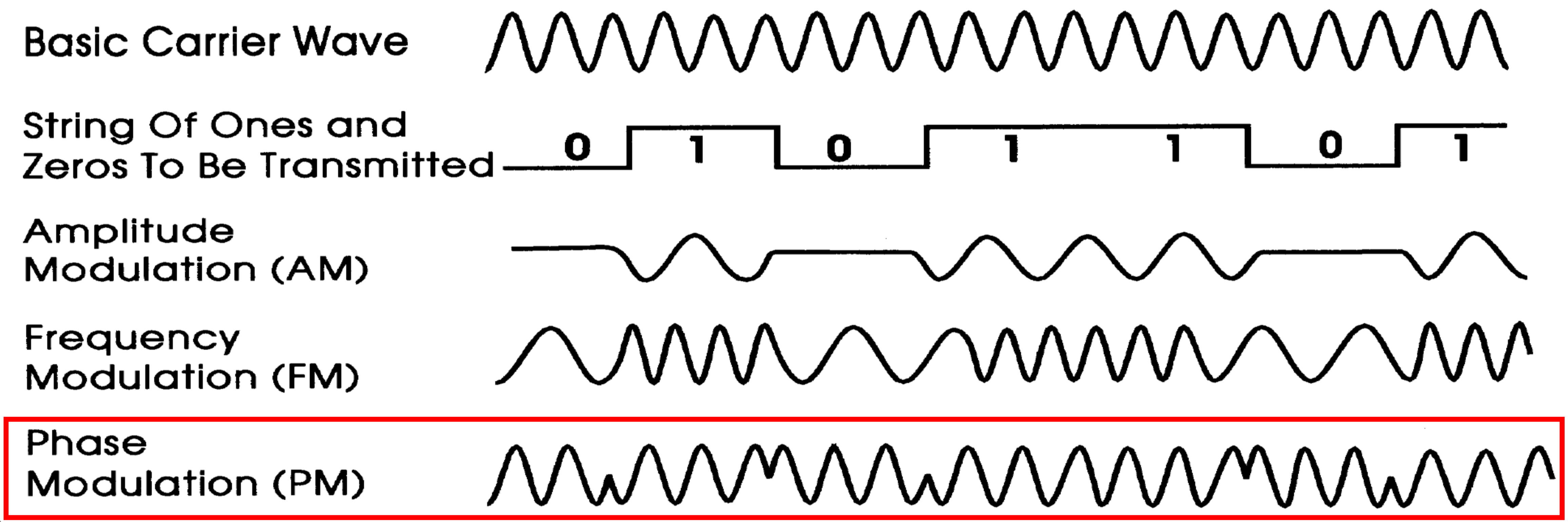


1 "Chip" ~1 Microsecond
= 300 Meters

Signal Processing Allows Resolution To ~ 1 % Of Chiplength

= 3 Meters

Basic Signal Structure

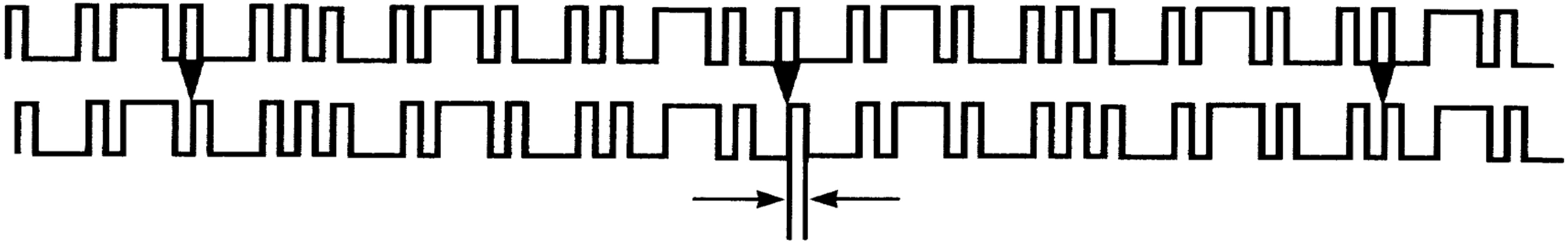


P Code

- 10 times faster than C/A code
- The code length is $2.3547 * 10^{14}$ bits
- Corresponding time span is 266.4 days
- Split into 38 segments
  - 32 are assigned to GPS satellites
  - Satellites often identified by which part of the message they are broadcasting
- Chip length is 30 m
- To protect against deliberate misinformation, combined with the encrypting W-code to produce Y-code
  - Only accessible if you know how to decode (*i.e.* military applications)

GPS Resolution - Code Phase

P- Code 10.23MHz

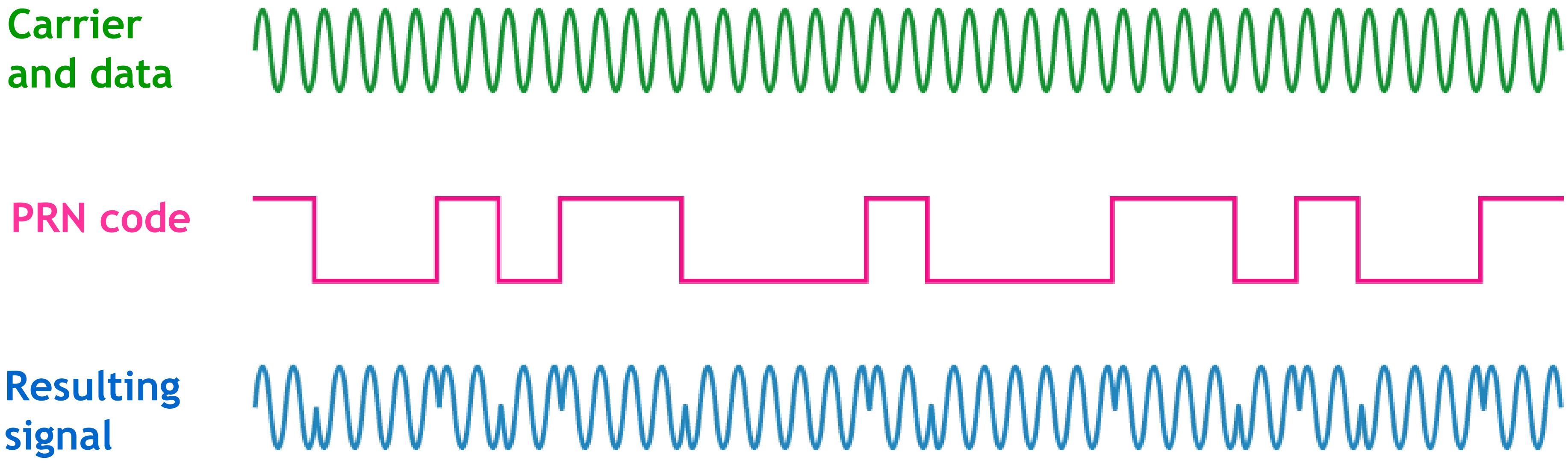


1 "Chip" ~1/10 Microsecond
= 3 Meters

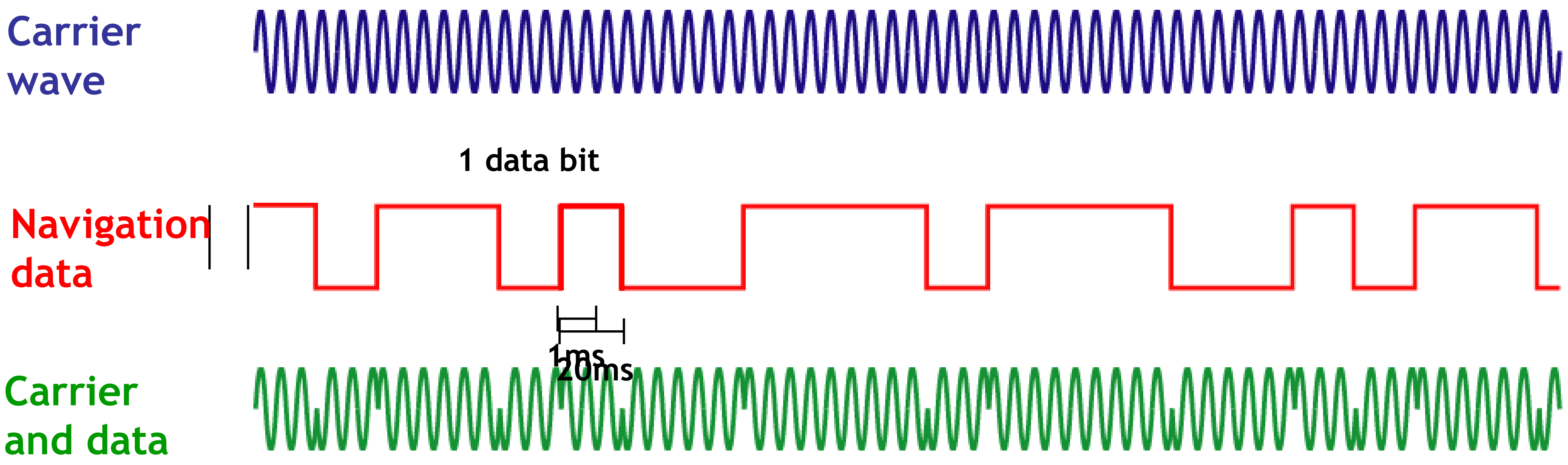
Signal Processing Allows Resolution To ~ 1 % Of Chiplength

= 0.3 Meters (30 Cm)

GPS signal Carrier Wave + Code data

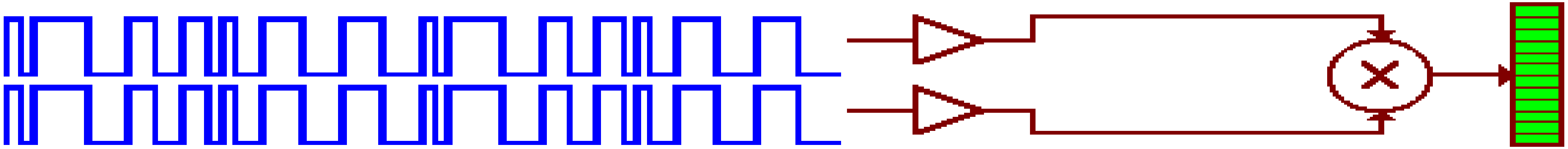


GPS signal carrier wave + Navigation data



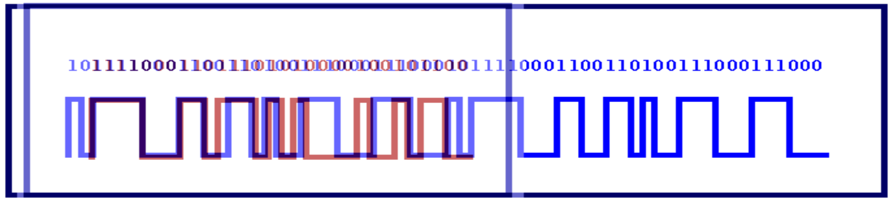
How is the Range measured using codes?

- Receiver slides replica of code in time until it finds correlation with the signal from satellite.



- If receiver applies different PRN code to SV

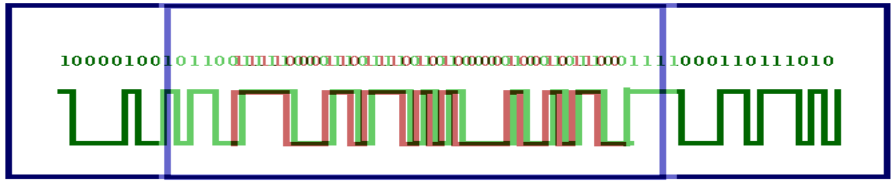
signal, then there will be no correlation.



No Correlation with a Different PRN Code

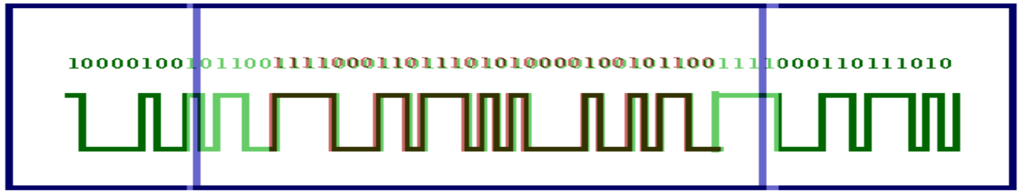
- when receiver uses same PRN code as SV and

codes begin to align and some signal power detected.



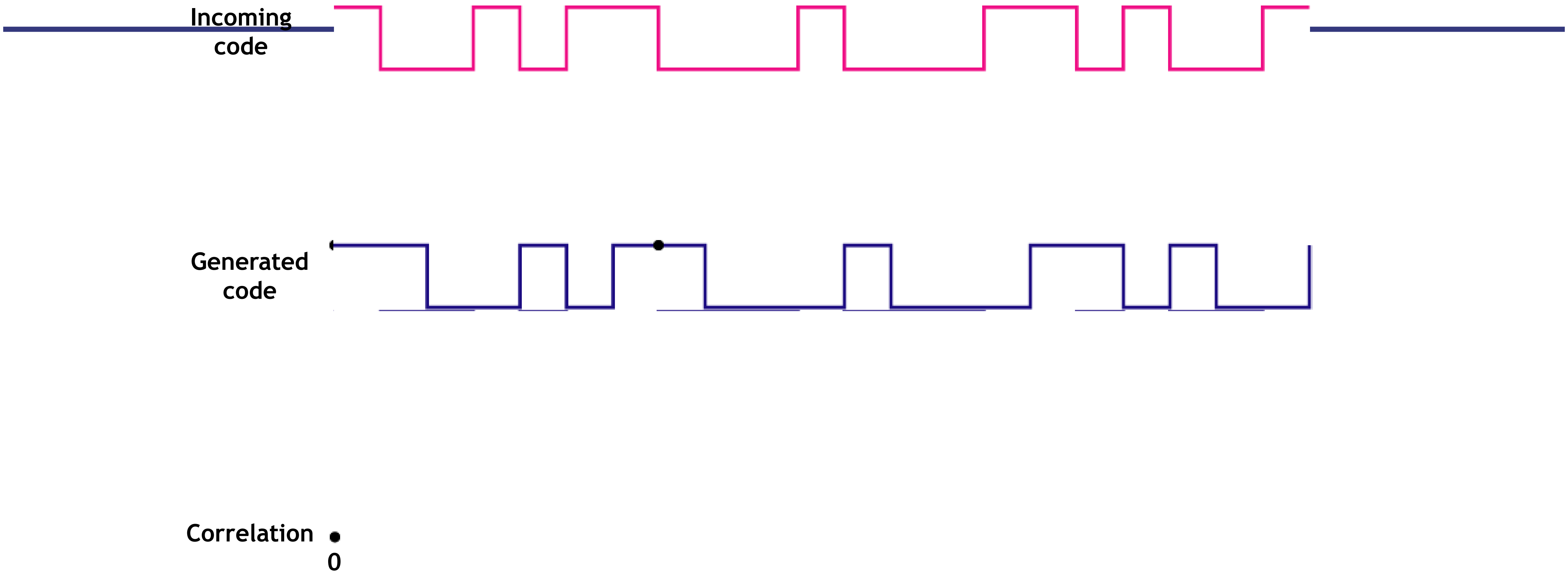
Partial Correlation of Identical Receiver and Satellite PRN Codes

- when receiver and SV codes align completely, then full signal power is detected.



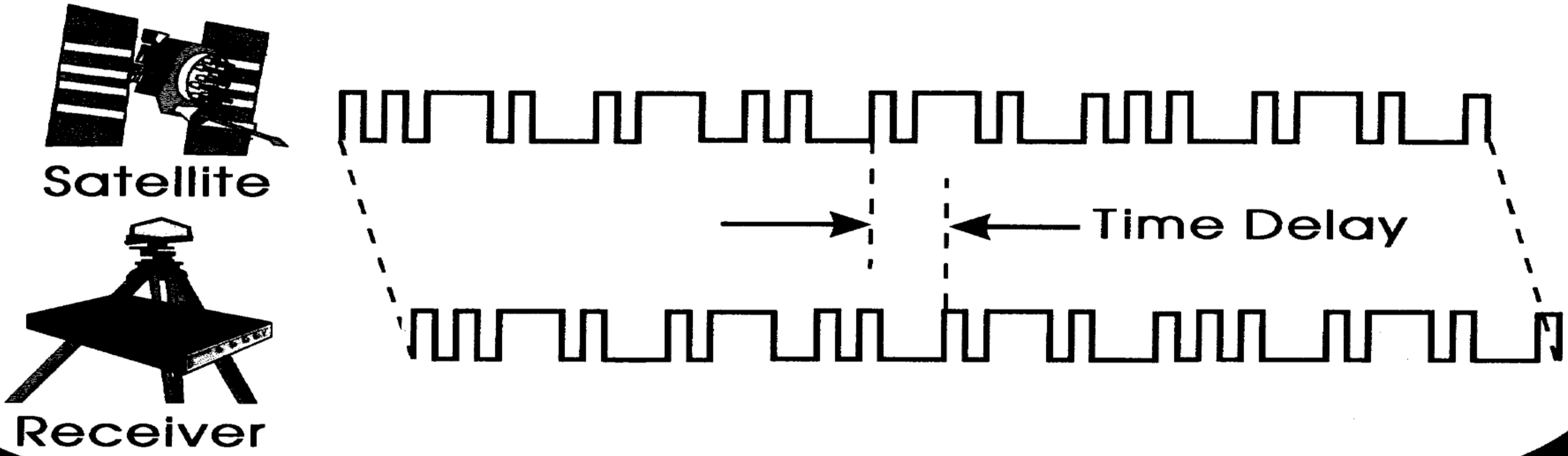
Full Correlation (Code-Phase Lock) of Receiver and Satellite PRN Codes

Acquisition

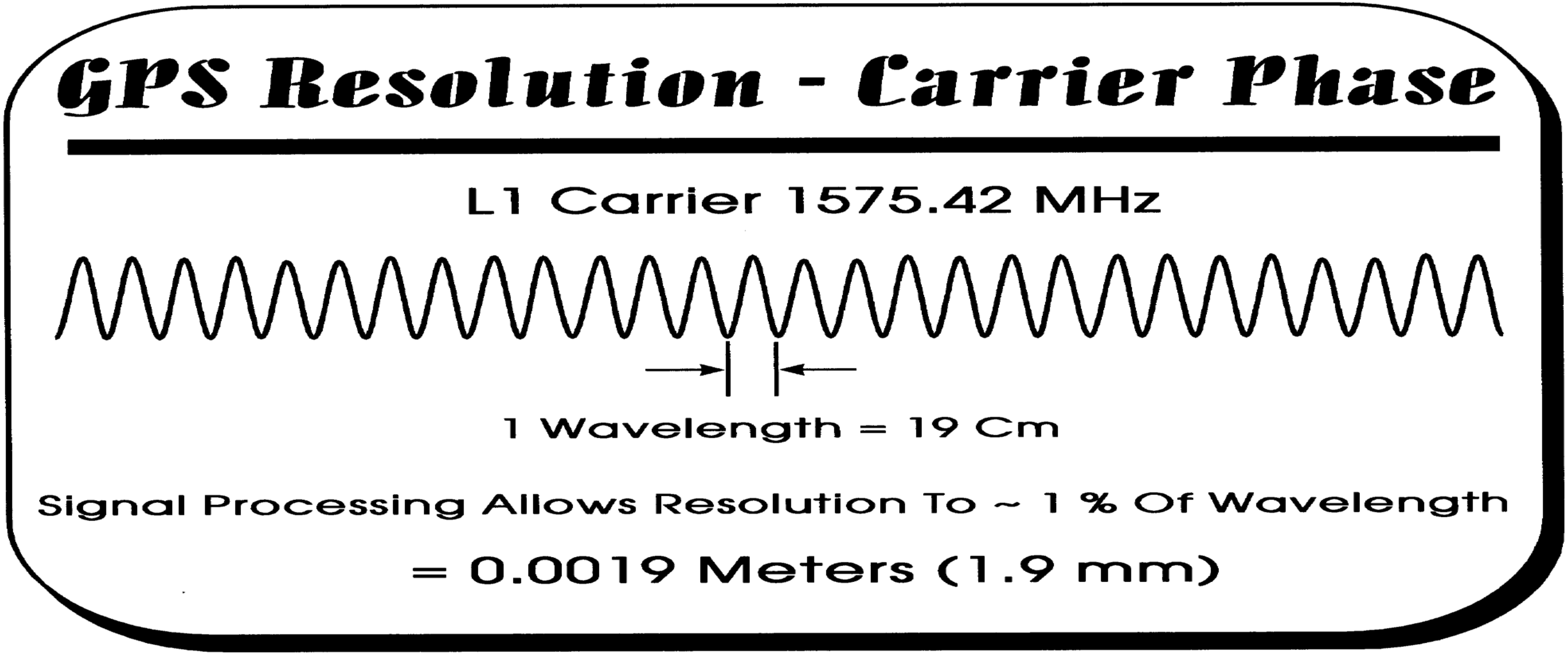


The Code Is The Key

- Receiver "Knows" Each SV′ s Unique Code
- Receiver Generates Replica Of Each Code Internally
- Receiver Then Measures The "Lag-Time" Of SV Code



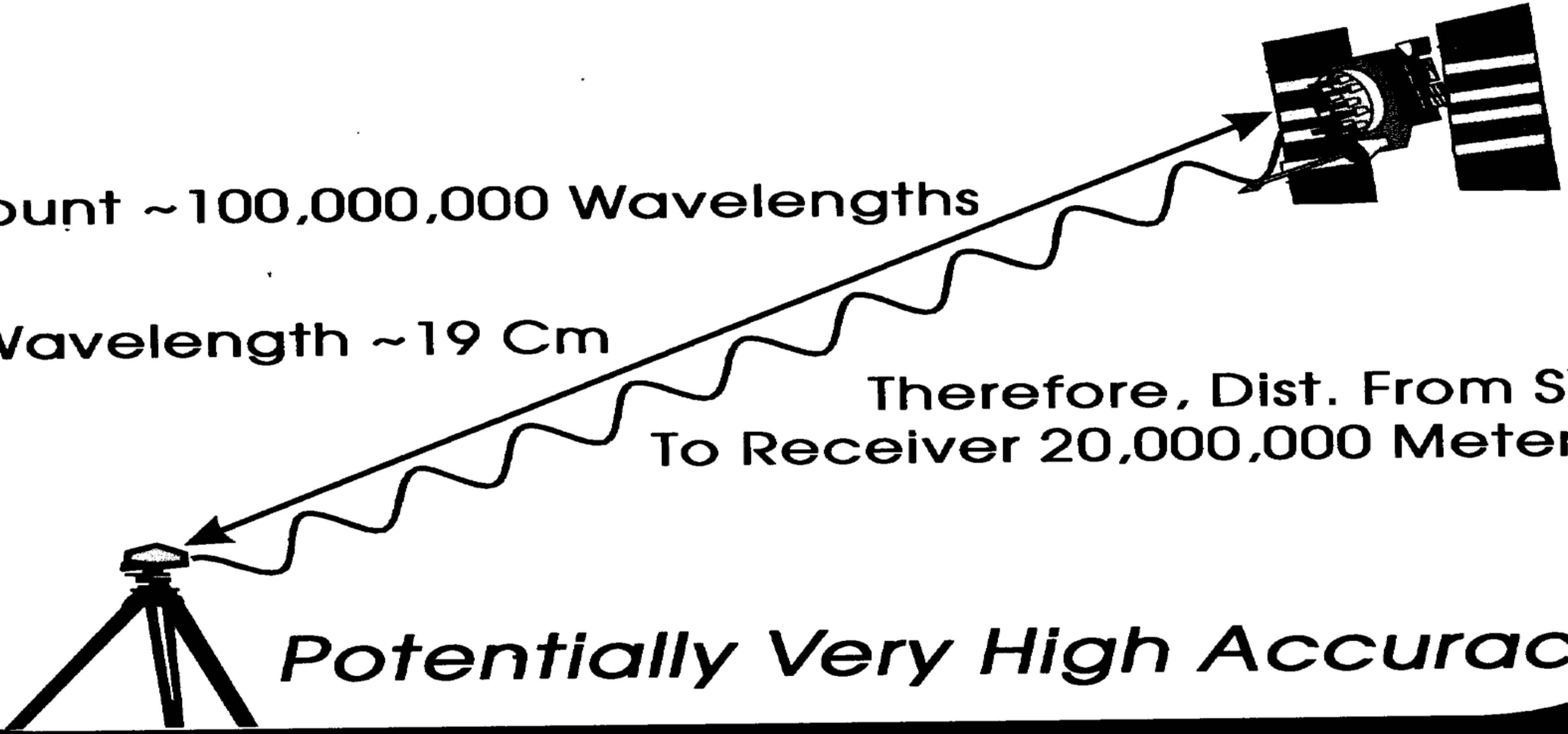
How is the Range measured using Carrier wave?



Carrier Phase Positioning

Higher Frequency Gives Higher Accuracy

Count Wavelengths Instead Of Chiplengths



Count ~100,000,000 Wavelengths

1 Wavelength ~19 Cm

Therefore, Dist. From SV
To Receiver 20,000,000 Meters

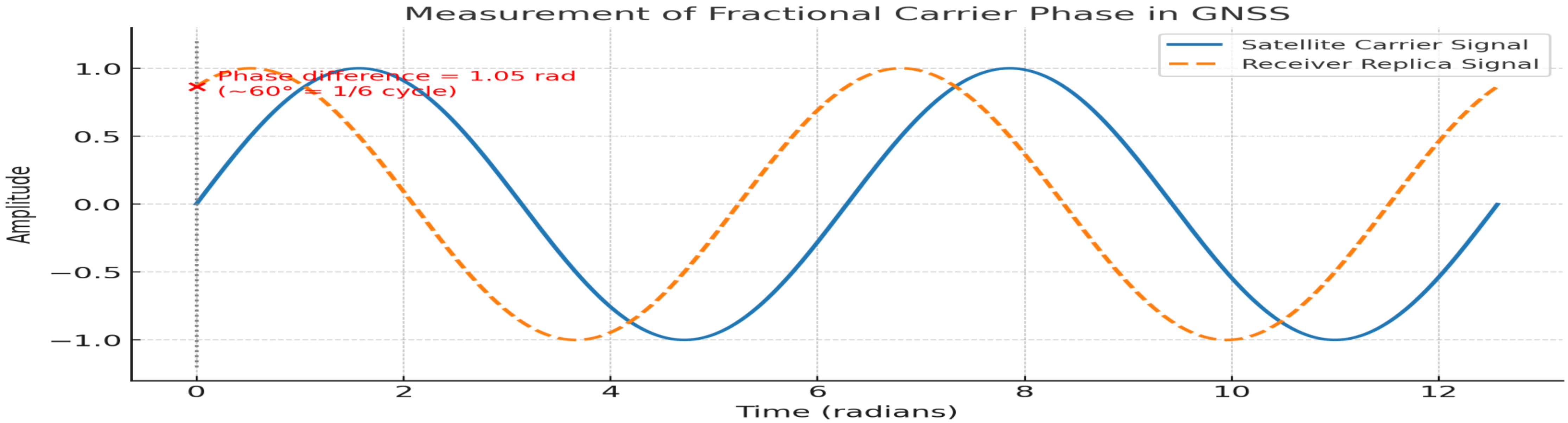
Potentially Very High Accuracy

But There is a problem….

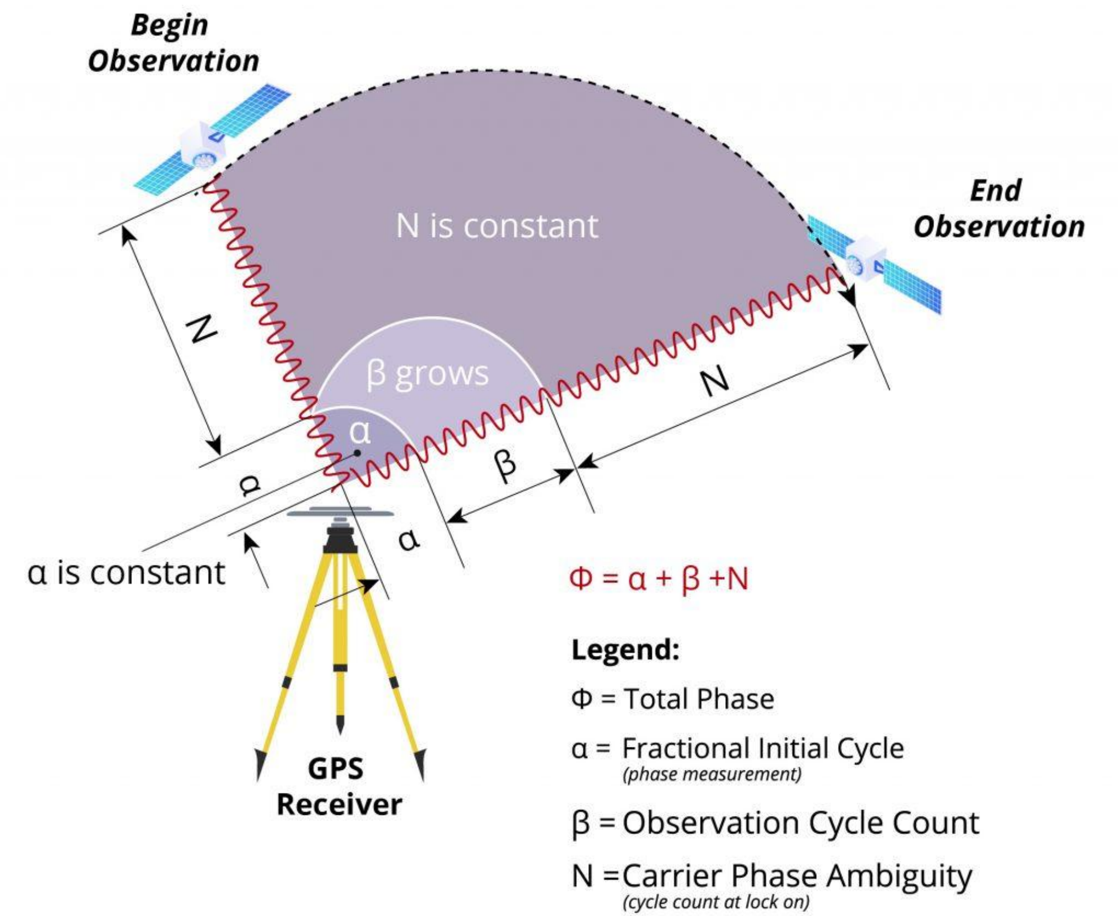
- Receiver can measure the phase of the carrier wave the moment it reaches the receiver, and it continues to measure the number of full cycles till the '**lock**' is maintained.
- Whereas we are interested in measuring the Number of full cycles the signal has travelled before reaching the receiver. Computing this unknown number of full cycles is called 'Ambiguity Resolving'.
- Only when ambiguity is resolved, the measured range is correct, and one can get accurate coordinates.

Carrier phase measurement technique

- In **carrier-phase GNSS measurements**, the receiver measures the phase of the carrier wave. But the phase only tells us the **fractional part** of the wavelength.
- We don't know how many **whole wavelengths (N)** are between the satellite and the receiver.This unknown integer N is called the **carrier-phase ambiguity**.



GNSS provides 3D positioning



N is constant because it's the unknown starting integer cycle count at lock-on.

α is constant because the receiver keeps track of fractional phase continuously.

β is the only part that grows with time.

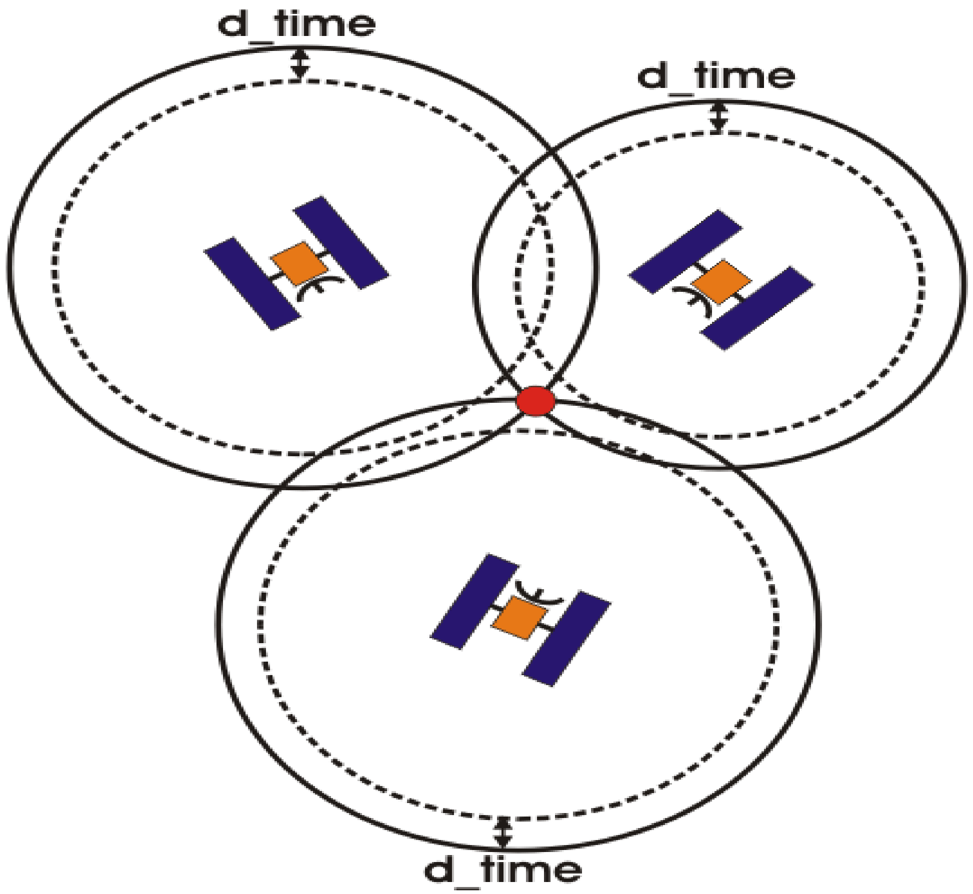
When does N or α change?
Cycle slip: if the receiver loses lock (due to obstruction, multipath, weak signal), then counting fails → a new lock begins → new N and α values appear.
Otherwise, during continuous tracking, **N and α remain constant**.

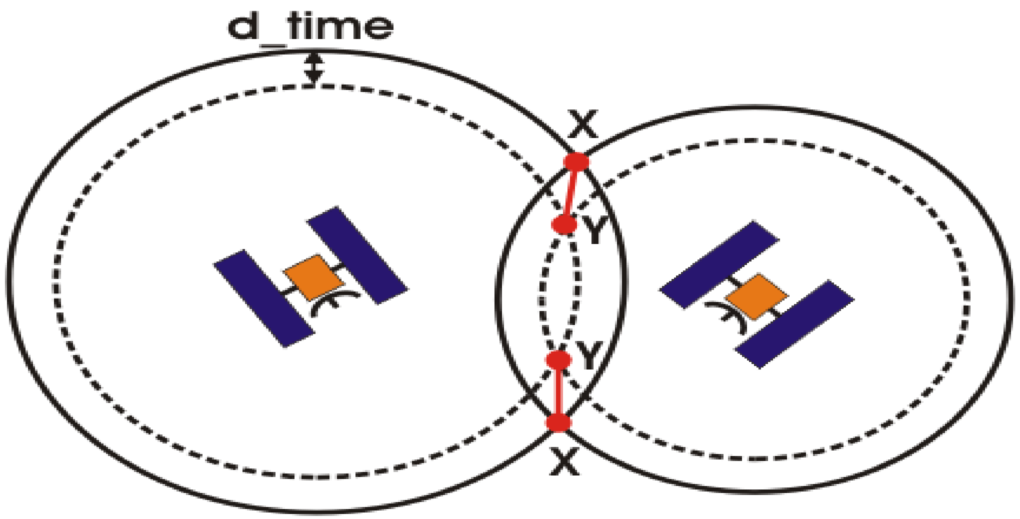
Space segment: Distance from satellite

- Distance = Velocity X Travel time
  - Difference in time from satellite to time received gives distance from satellite
- Satellites have very accurate Rubidium Atomic clocks on board and very accurate ephemeris information
- Radio waves travel with speed of light
- Light speed = 300,000 km/second
  - Out of sync by 1/100th of second equals error of 3000 km!
  - Receivers should have nanosecond accuracy (0.000000001 second) to get 30cm accuracy!

But wait!

There's a caveat to this. YOU don't carry an atomic clock. They are expensive and heavy. YOUR receiver clock is going to be off by some random amount when it compares time with the satellites.



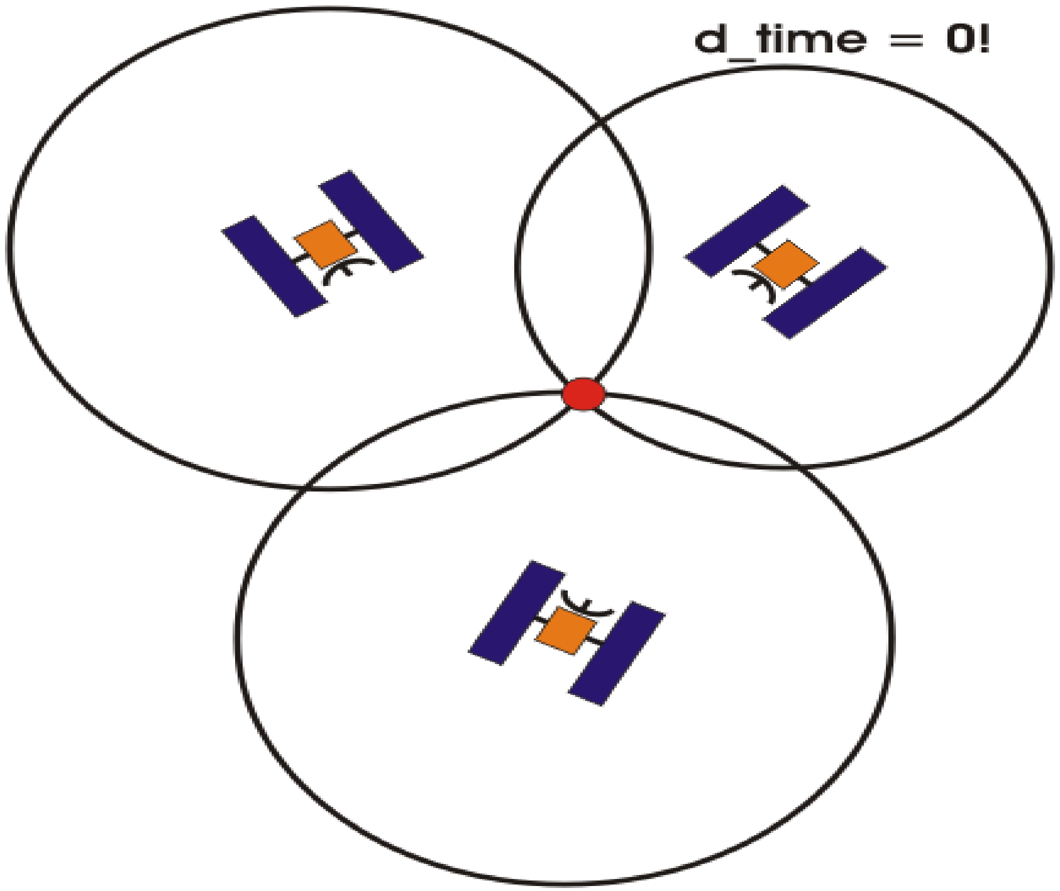


<u>How do we get around this?</u>

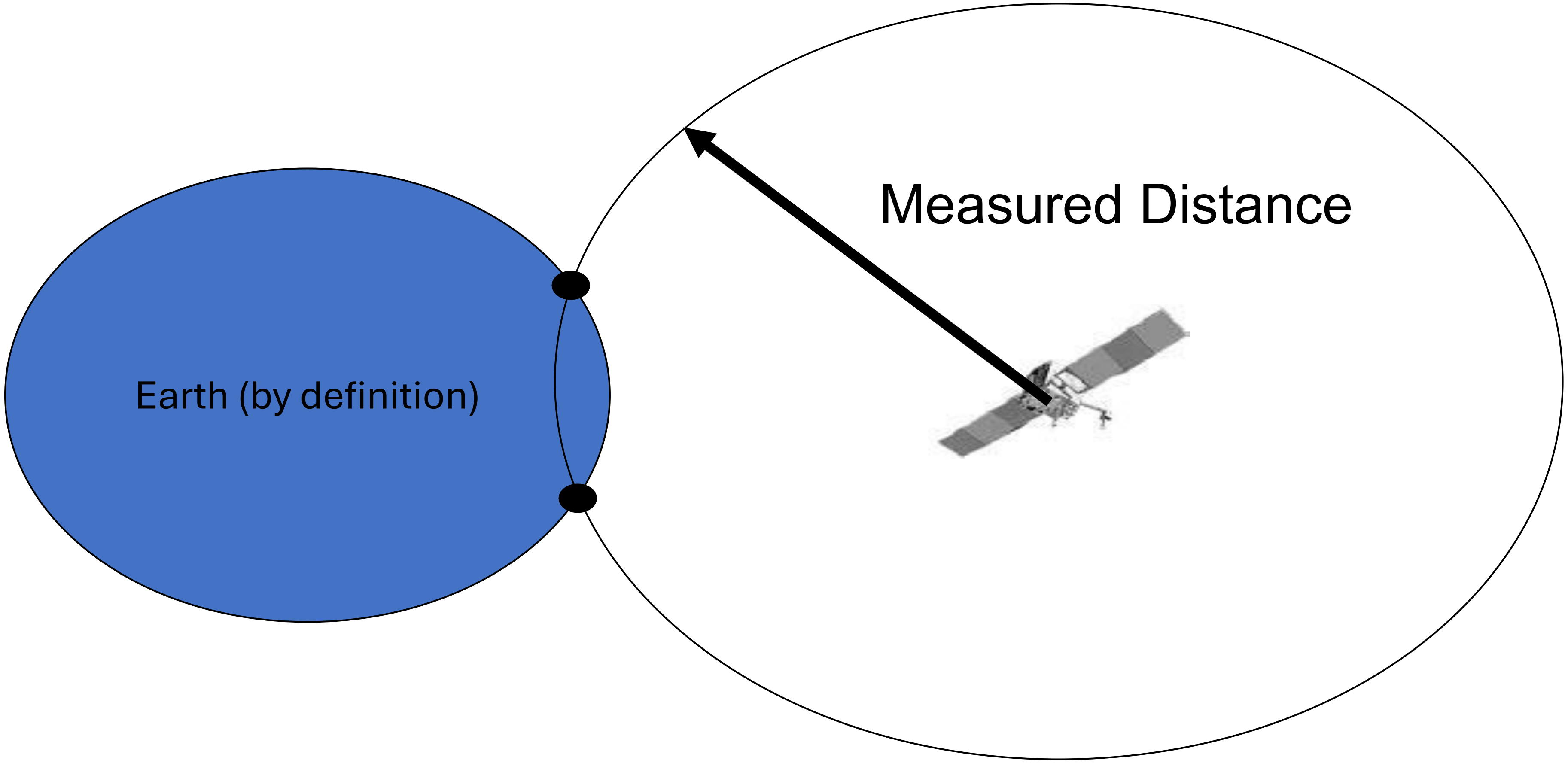
An atomic clock in your pocket

We add a 4th measurement!

- If we *knew* the time, a 4th satellite would be redundant.
- However, the extra satellite can ONLY match up with the other three at the *CORRECT* time. It removes the error and calibrates your receiver at the same time!!!
- This technique lets your GPS unit know the time to within ~100 ns (0.0000001 seconds).



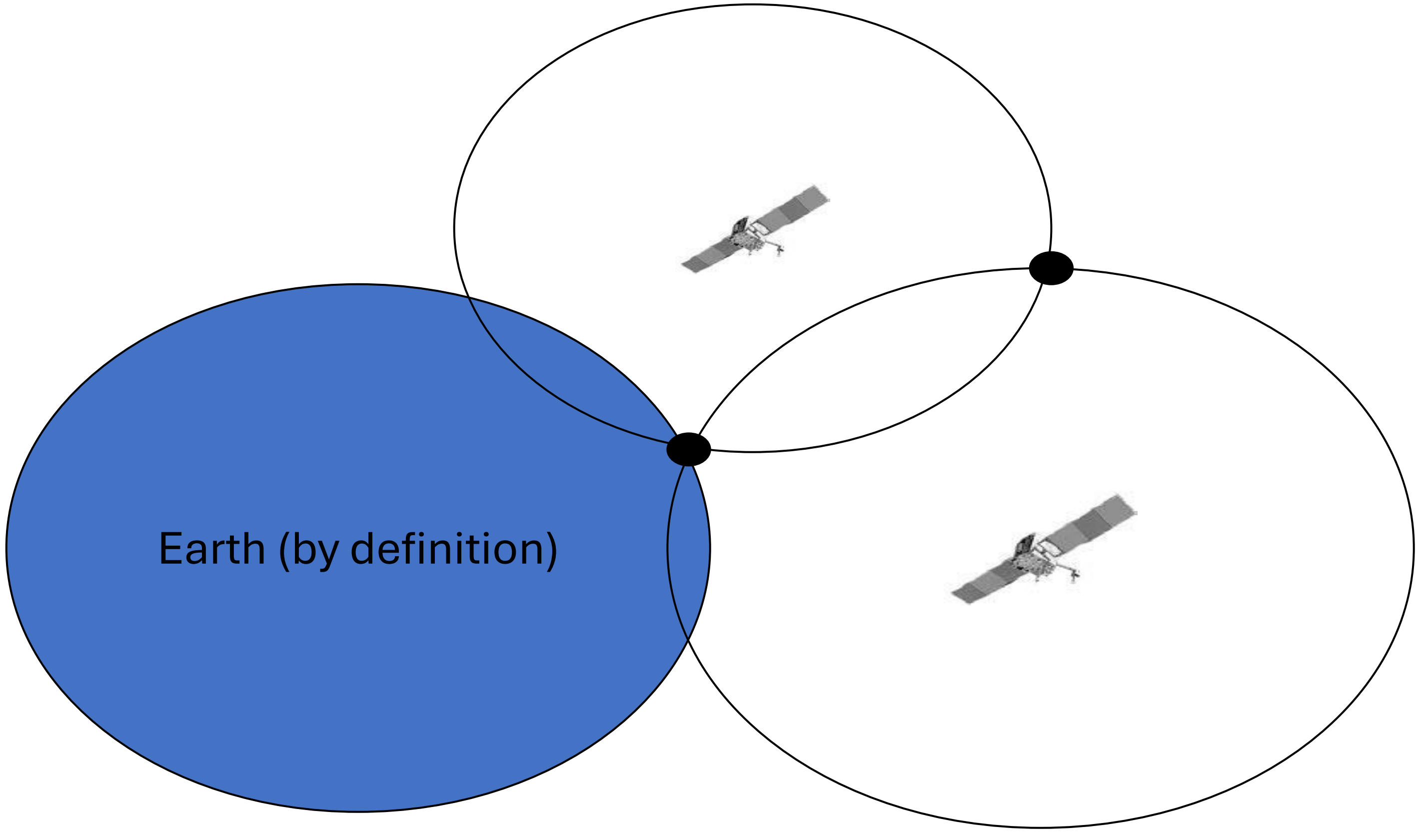
The satellite knows where it is.



We know the distance from the satellite by the code correlation.

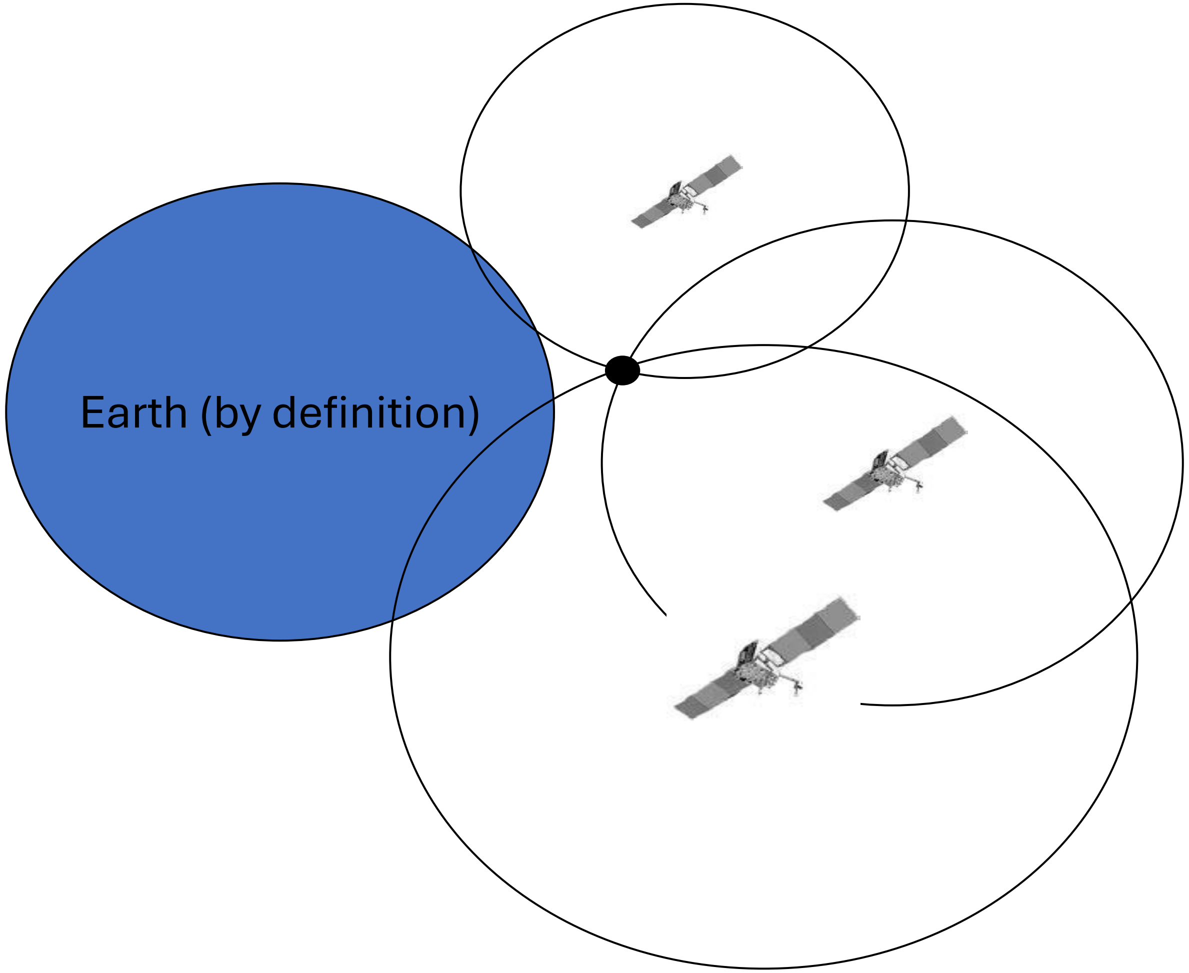
So we know where we are on a big circle (sphere) around the satellite.

Add another satellite



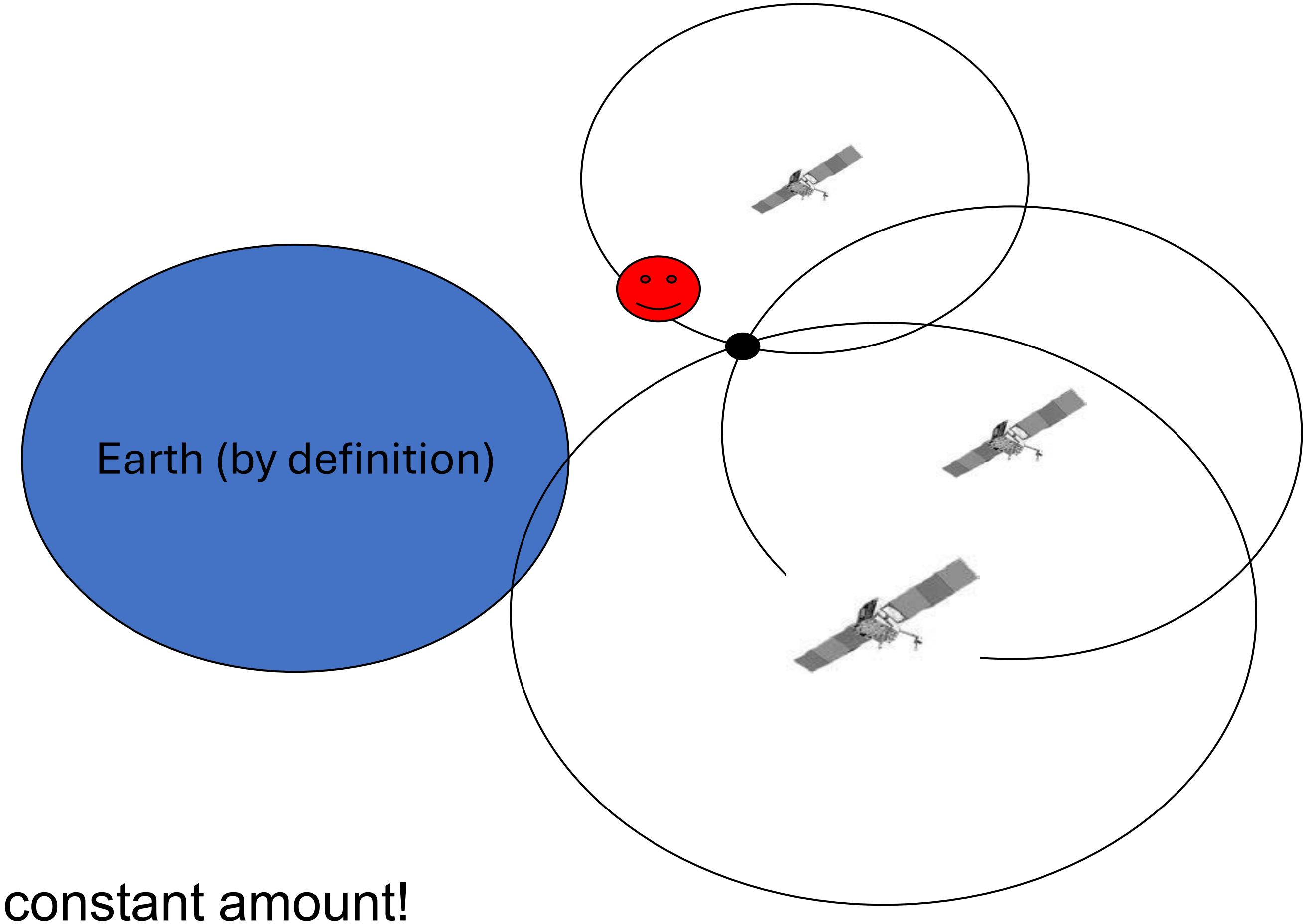
Two dimensional example:
We're in one of two spots.

Add another satellite



Again: Two dimensions – 3 satellites – we know where we are!

Remember the pesky clock problem?

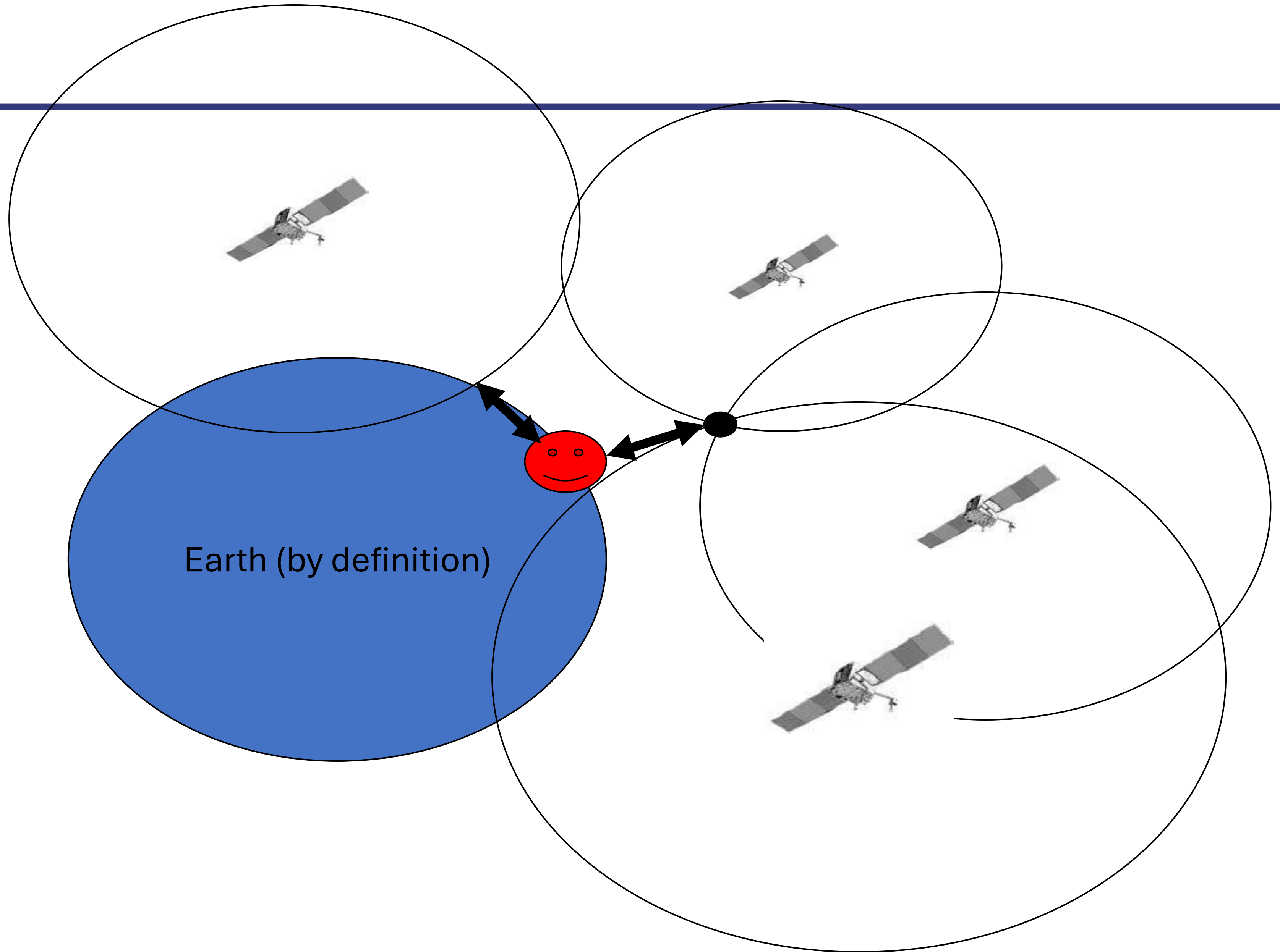


Satellites have expensive clocks.
Our receiver doesn't!

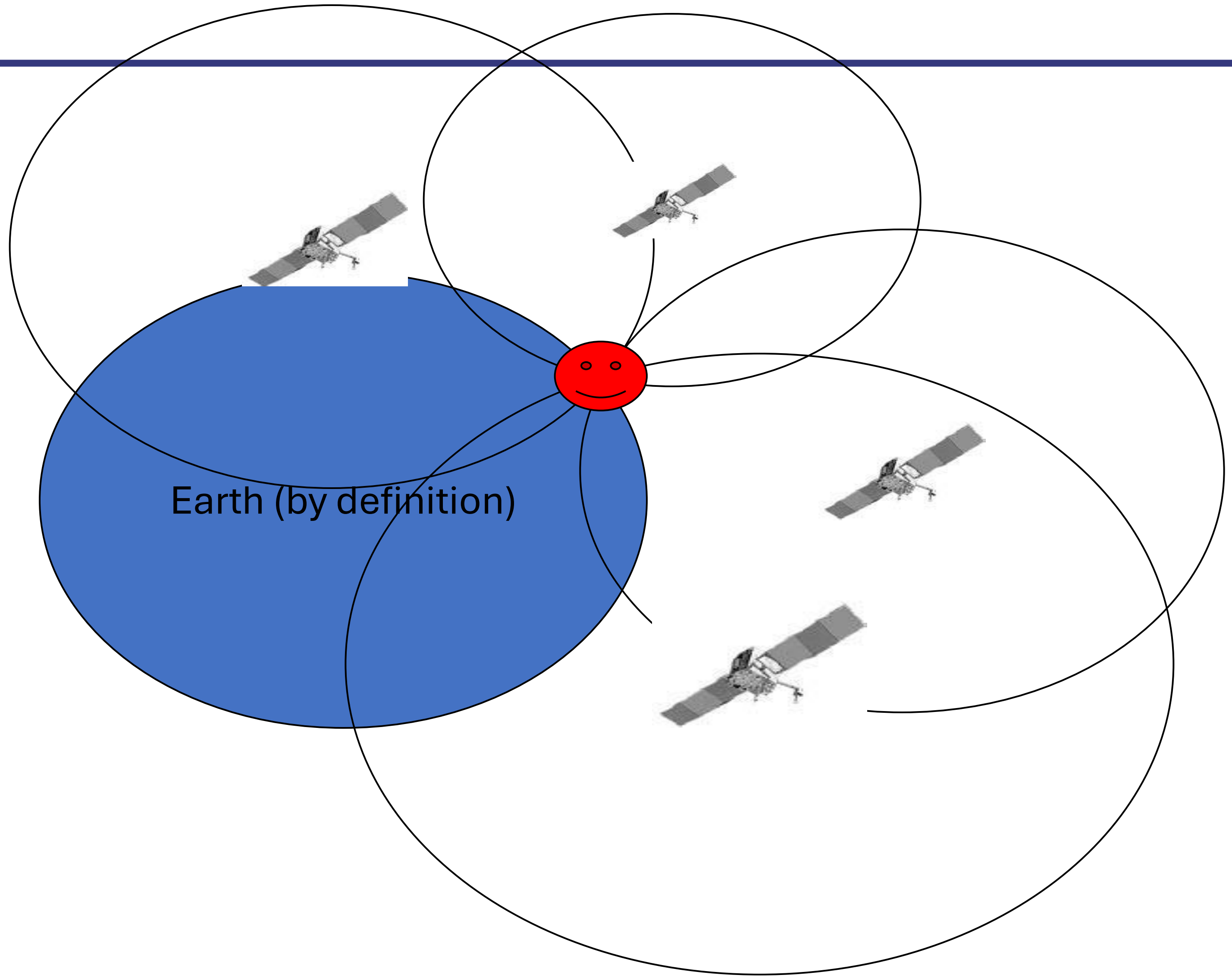
Our clock is "off".
So our distance is off – but by a constant amount!

Old trick: Add another satellite

What number do we add or subtract from the time correlation to make everything come together?



Add or subtract the time offset number



Now you got the time.

QUESTIONS?

THANKS